Selection of Time Headway for String-Stable CACC under Noisy V2V Communication

The base paper reproduced to its printed digits, then extended — validated across three independent simulation backends

Control Systems — Project Presentation

Base paper: G. Ma, P. R. Pagilla, S. Darbha, “Selection of Time Headway in Connected and Autonomous Vehicle Platoons Under Noisy V2V Communication,” IEEE Transactions on Intelligent Transportation Systems, vol. 26, no. 1, pp. 1029–1038, January 2025. doi: 10.1109/TITS.2024.3498701

Backends: Python scientific core · ROS 2 distributed (one process per vehicle) · MATLAB / Simulink block model

Status: reproduced to the paper's printed digits · 87 unit tests green · Monte-Carlo certification run CERTIFIED

TEAM

Project Team

Control Systems — Selection of Time Headway for String-Stable CACC under Noisy V2V Communication

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Problem statement — the physical problem

WHY PLATOONING

- **Highway capacity is spacing-limited, not speed-limited**
  - capacity ≈ v / (L + r + h·v) vehicles per lane per second
  - at v = 25 m/s: every 0.1 s of headway costs 2.5 m per vehicle
  - humans need h ≈ 1.5–2 s; standard ACC still needs h ≥ 1 s
- **Two failure modes — only one is obvious**
  - individual: does vehicle i track its predecessor? Solved by any stable feedback loop.
  - **STRING stability: does a disturbance GROW as it travels backwards down the platoon?**
- **String instability is what phantom traffic jams are**
  - growth is geometric, g^N: a per-hop gain g = 1.0035 is invisible over 11 vehicles (1.039×) and a factor 2 over 200
  - **nothing in a 200 s time trace reveals it — the verdict must be frequency-domain**

WHY V2V CHANGES THE PROBLEM

- **Radar alone cannot close the gap**
  - predecessor-following at CONSTANT SPACING is provably string-unstable: the double integrator forces $\sup|\Gamma| > 1$
  - constant time headway d = r + h·v fixes it — but the price is exactly the road space we wanted back
- **V2V feedforward is the lever**
  - broadcasting the predecessor's ACCELERATION lets the follower act BEFORE the gap changes
  - **in this model it halves the headway floor: $2\tau_o \rightarrow \tau_o$**
- **…but the lever is a radio**
  - **the one signal CACC adds over ACC is the one that travels over an imperfect channel**
  - **design for a clean channel, deploy on a dirty one → SILENT string instability**

A 12-vehicle platoon at h = 0.95 s occupies 345 m; at h = 0.88 s it occupies 324 m — 1.75 m per vehicle, ≈6 % more flow.

Problem statement — the research problem

From "does the paper work?" to "would it survive deployment?"

THE QUESTION THE BASE PAPER ASKS

- **Given a signal-to-noise ratio ρ on the V2V link carrying the predecessor's acceleration, what is the SMALLEST time headway h_w at which the platoon is still GUARANTEED string stable — for every admissible channel realisation and every vehicle lag $\tau \in (0, \tau_0]$?**
- Its answer is a closed form: the bound $h_{w,lb}(k_a, \rho)$, the feedforward gain k_a^* that minimises it, and the resulting minimum $h^*_{w,lb}$.

WHAT ITS ANSWER ASSUMES — each assumption is a deployment gap

- channel statistics are TIME-INVARIANT and known at design time
- the guarantee holds in the MEAN sense and is BINARY (pass / fail)
- there is NO communication delay and NO packet loss
- every link is HONEST; only the radio is noisy, the radar is clean
- the gains are chosen ONCE, offline, for one value of ρ

THE THREE QUESTIONS WE ADD

- **1. Real channels vary. What happens when ρ changes DURING operation, and can the platoon ESTIMATE it and adapt online?**
- **2. Can a binary worst-case verdict be made QUANTITATIVE — a calibrated probability of exceedance instead of a bound paying for 10^{-6} events?**
- **3. What if a link LIES? A predecessor-following chain has no redundancy, so consensus-style filtering cannot be applied.**

Success criteria: reproduce the paper exactly · quantify its deployment gaps · build and validate the fix · certify on a Monte-Carlo harness · agree across three independent backends.

Base paper — model, spacing policy, control law, noisy channel

$N = 12$ followers · $\tau_o = 0.5$ s · $d = 5$ m · $\rho = 5$ (13.98 dB) · $v_{ss} = 25$ m/s · $a_o(t) = 0.5 \sin(0.1(t-10))$ over one period

Vehicle dynamics, eq. (1) — double integrator + actuator lag

$$\ddot{x}_i(t) = a_i(t), \qquad \tau\,\dot{a}_i(t) + a_i(t) = u_i(t), \quad i = 1 \dots N$$
$$\tau \text{ UNCERTAIN: } \tau \in (0, \tau_o],\ \tau_o \text{ known} \implies G(s) = 1/(s^2(\tau s+1))$$

τ is the DOMINANT lag of the inner acceleration loop, not one component's time constant. Heterogeneity is absorbed by $\tau_o = \max_i \tau_i$, so a design certified on $(0, \tau_o]$ covers any mixed fleet below τ_o.

Spacing errors — Definition 1 and eq. (2)

$$e_i = x_i - x_{i-1} + d \qquad \text{(constant-spacing error)}$$
$$\delta_i = e_i + h_w\, v_i \qquad \text{(CTHP error)} \qquad (2)$$

CTHP control law, eq. (6) — three terms, three information sources

$$u_i = k_a \cdot w_{i,i-1}(t) \cdot a_{i-1} - k_v(v_i - v_{i-1}) - k_p \cdot \delta_i$$

term 1	FEEDFORWARD	V2V RADIO	pre-empt the manoeuvre
term 2	DAMPING	RADAR	kill the speed error
term 3	SPACING	RADAR + IMU	regulate the gap

Only the feedforward term is corrupted, because only it travels over the radio. That asymmetry is the entire premise of the paper.

The 16-bit multiplicative channel, eq. (5), and the SNR, eq. (4)

$$w_{i,i-1}(t) = (1 - 1/\rho) + (1/\rho) \cdot \Sigma_{j=0}^{15}\, z_{i,j}(t) \cdot 2^{-j}$$
$$z_{i,j} \sim \text{Bernoulli}(\gamma_{i,j}) \text{ independent}; \quad \rho = 10^{\text{SNR}/20}$$
$$\implies \text{support STRICTLY BOUNDED: } 1 - 1/\rho \le w < 1 + 1/\rho$$
$$\implies E[w] = 1.0482 \neq 1 \quad \text{(the channel is BIASED optimistic by 4.8 \%)}$$
$$\implies \text{effective gain } \tilde{k}_a := k_a \cdot E[w] = 0.5241 \qquad (12)$$
$$\gamma_{i,j} \text{ UNKNOWN} \implies \text{robustify over } I = [0.4, 0.6]$$

Bounded support is what makes a DETERMINISTIC guarantee possible at all — a Gaussian noise model has tails, so no finite headway could guarantee string stability.

Base paper — Theorem 2, the closed forms, and the five cases

Eqs. (14), (16)–(19) and §IV — the whole result set in one page

Spacing-error propagation, eq. (14): $\delta_i(s) = \tilde{H}(s;\tau)\,\delta_{i-1}(s)$

$$\tilde{H}(s;\tau) = \frac{\tilde{k}_a s^2 + k_v s + k_p}{\tau s^3 + s^2 + \gamma s + k_p}$$

where $\gamma := k_v + h_w k_p$ — headway enters ONLY here, and ONLY in the denominator $D(s)$

ROBUST STRING STABILITY (Def. 2): $\|\tilde{H}(j\omega;\tau)\|_\infty \le 1$
for ALL $\tau \in (0, \tau_o]$ AND ALL $\tilde{k}_a \in I$, plus internal stability

Theorem 2 — the three results the paper exists to produce

(b) given $0 < k_a < 1/(1 + 1/\rho)$ and $h_w > h_{w,lb}$, (16)

where $h_{w,lb} := 2\tau_o \cdot [\,1 - (1 - 1/\rho)\,k_a\,] \,/\, [\,1 - (1 + 1/\rho)^2 k_a^2\,]$ (17)

there EXIST $k_v, k_p > 0$ with $\|\tilde{H}\|_\infty \le 1\ \forall\ \tau \in (0, \tau_o]$

(c) $k_a^* = [\,(1 - 1/\sqrt{\rho}) / (1 + 1/\sqrt{\rho})\,] \cdot [\,1 / (1 + 1/\rho)\,]$ (18)

$h^*_{w,lb} = \tau_o \cdot (1 + 1/\sqrt{\rho})^2 / (1 + 1/\rho)$ (19)

The five numerical cases of §IV

1	k_a=0.5, h_w=0.95, k_v=0.63, k_p=0.009	$h_w > h_{1b} \Longrightarrow$ STABLE
2	same gains, h_w = 0.65 s	$h_w < h_{1b} \Longrightarrow$ UNSTABLE
3	$k_a = k_a^*$, h_w = 0.88, k_v=0.85, k_p=0.003	at the optimum
4	Case 1 vs Case 3	platoon 345 m vs 324 m
5	Case 1 gains, heterogeneous τ_i	the $\tau \in (0,\tau_o]$ claim holds

At ρ = 5, τ_o = 0.5 s the paper prints k_a bound 0.8333, $h_{w,lb}(0.5)$ = 0.9375 s, k_a^* = 0.3183 and $h^*_{w,lb}$ = 0.8727 s — all four reproduced to 0.00 % (see Results 1 / 5). Case 2 is the tightness test: the bound is violated by only 6 % in headway and the spacing errors GROW down the string.

Novelty, phase-wise — the seven phases

★ = core research contribution ★★ = the flagship extensions

Phase	What it adds	Assumption removed	Key result	ADR
0 Reproduce	Full re-implementation from the paper: CTHP law, 16-bit channel, RK4 platoon, frequency-domain analysis	— (fidelity baseline)	**printed digits exact: 0.9375 / 0.3183 / 0.8727 / 0.8333**	D-001…07
1 Gap audit	V2V latency θ on the feedforward path + Bernoulli packet loss with ZOH, at FIXED case-A gains	"no delay", "no packet loss"	**DELAY binds, not noise: only ≈0.2 s of budget at the paper's own design**	D-008
2 Distribute	One OS process per vehicle on ROS 2; wall-clock substepped integration; radar extrapolation by message age	"a monolithic simulator is enough"	**< 2 % per-hop agreement with the offline core under real OS jitter**	D-011…15
3 Adapt ★	Online SNR estimation $\hat{\rho}$ from radar-vs-beacon residuals → rate-limited headway adaptation + timestamp predictor	"$\gamma_{i,j}$ are time-invariant"	**$\hat{\rho}$ median {10.4, 3.1, 10.7} vs true {10, 3, 10}; delay penalty flatlined**	D-016
4 Re-tune	Online gain scheduling: hold γ at 90 % of the (28a) ceiling for the estimated ρ	"gains are fixed once chosen"	**breaks the fixed-gain wall at $\rho^* \approx 1.8$: $\|\tilde{H}\|_\infty$ 1.019 → 0.99999**	D-017
5 Flagship ★★	A: spatial ρ(x) map with look-ahead · B: physics-consistency V2V trust gate · C: exact channel law → chance-constrained headway	"reactive is enough", "links are honest", "binary worst case"	**A: 0 % unstable in-zone (vs 6.9 %) · B: collision → 8.4 m gap · C: −10.5 % headway at 1e−2 risk**	D-021…23
6 Certify	Monte-Carlo scenario×seed harness → PASS/FAIL report; plus Simulink as an independent third backend	"one implementation is evidence"	**CERTIFIED across 4 scenario families; Simulink ↔ Python agree to 1e−4**	D-018, 24

Systems — what we built, and the three backends it runs on

THE MODEL AND THE CHANNEL

S1 VEHICLE + CONTROLLERS vehicle.py, controllers.py

- state $x = [p, v, a]$; $\dot{a} = (u - a)/\tau$, u saturated to $[-8, +3]$ m/s² BEFORE integration
- $\tau = 0.5$ s, $L = 4$ m, $r = 1$ m $\Longrightarrow$ $d = 5$ m, matching the paper
- three laws behind ONE interface — ACC (radar only), CACC (+ unity feedforward), CTHP (the base-paper law)
- **CTHP is STATELESS, so the gain scheduler can swap the instance mid-run with no state transfer**

S2 V2V CHANNEL network.py, qos_map.py

- four independently switchable impairments — NOISE (w per eq. 5, held 10 ms) · DELAY θ · LOSS + zero-order hold · ATTACK injection on one link
- **the ρ-INDEPENDENT bit cache: it stores $U' = \Sigma\, z_j\, 2^{-j}$, not w, so a ρ-schedule RESCALES the same realisation instead of redrawing it**
- **$\Longrightarrow$ two zone studies are seed-comparable: the difference between runs is the schedule, not the dice**
- QoSMap gives channel quality by POSITION, ρ(x), and exposes it at (x + v·preview) — the look-ahead that makes adaptation predictive rather than reactive

ANALYSIS, ADAPTATION, AND EXECUTION

S3 ANALYSIS + CERTIFICATION analysis.py, certificate.py

- **delays evaluated EXACTLY as $e^{-j\omega\theta}$, never Padé — margins here are 10^{-3} and Padé phase error would corrupt exactly the verdicts we need**
- **min_stable_headway BISECTS on h, because (28a)/(28b) are sufficient-not-necessary; hinf_worst grids $\tilde{k}_a$ across I so the binding endpoint is CHECKED, not assumed**
- ChannelLaw.build(ρ) — the EXACT law of w, $2^{16} = 65\,536$ atoms

S4 ADAPTATION estimation.py, trust.py

- **$\hat{\rho}$ from radar-vs-beacon residuals over an AGE-based window — a count-based window was tried first and FAILED ($\hat{\rho}$ stuck at 3.1 after zone exit)**
- h-table built ONCE by bisection, then rate-limited to 0.05 s/s; scheduler holds γ at 90 % of the (28a) ceiling

S5 THREE BACKENDS, ONE SET OF EQUATIONS

- **Python core — hand-written RK4 over the stacked state**
- **ROS 2 — one OS process per vehicle; the nodes IMPORT cacc.*, so there is no second implementation to drift**
- **Simulink — a width-N model built programmatically from a reviewable .m, so the .slx is a diffable artefact**

Declarative YAML scenarios $\Longrightarrow$ a study is a name plus a seed; each run writes results/<study>/<stamp>/ with figures AND metrics.json, so every number on a later slide traces to a file.

The maths in one page — from a car to a closed-form headway

Eqs. (1), (14), (17)–(19) and (28) — the chain, each step as a result rather than a working

GOAL

The four steps that matter. Every cancellation, sanity check and full working: report/PRESENTATION_GUIDE.md.

1 **THE CAR, AND THE GAP IT KEEPS.**

$$\tau\,\dot{a}_i + a_i = u_i \implies G(s) = \frac{1}{s^2(\tau s+1)} \quad \text{two poles at } 0$$

$$\text{gap} \quad d_i = r + h_w v_i \implies H(s) = h_w s + 1, \ |H(j\omega)| \geq 1 \ \forall\omega$$

⟹ **The spacing policy is secretly a low-pass filter. It costs no control effort and no fuel — it costs METRES, and that is the only currency this project spends.**

2 **ONE FUNCTION GOVERNS EVERY HOP.**

$$\tilde{H}(s) = \frac{\tilde{k}_a s^2 + k_v s + k_p}{\tau s^3 + s^2 + \gamma s + k_p} \qquad (14)$$

$$\text{with} \quad \gamma := k_v + h_w k_p$$

⟹ **Headway enters ONLY through γ, and γ ONLY in the denominator: it can add damping, never gain. A 200-car question is now a one-car question.**

3 **THE VERDICT — AND WHY IT CAN NEVER BE BEATEN.**

$$\text{gains MULTIPLY down a chain:} \quad |\tilde{H}|^N \implies \text{the test must be}$$

$$\|\tilde{H}\|_\infty = \sup_\omega |\tilde{H}(j\omega)| \leq 1 \quad \text{(no frequency gains energy)}$$

$$\text{but } \tilde{H}(0) = k_p/k_p = 1 \implies \|\tilde{H}\|_\infty < 1 \text{ is IMPOSSIBLE}$$

⟹ **The best achievable is exactly 1, so the low-frequency margin is ZERO by construction — which is why slow channel noise is the dangerous kind.**

4 **TWO CONDITIONS ON THE GAINS, THEN THE ANSWER.**

$$(28a)\ \gamma \leq \frac{[1 - (1+1/\rho)^2 k_a^2]}{(2\tau_o)} \quad \leftarrow \text{NO } h_w \text{ anywhere in it}$$

$$(28b)\ \gamma \geq \sqrt{2k_p[1 - (1-1/\rho)k_a] + k_v^2}$$

$$\text{chain} \implies h_w \geq \frac{2\tau_o(1-m\,k_a)}{(1-n\,k_a^2)}\ (17), \ \text{best } k_a \to (18),(19)$$

⟹ **At ρ = 5, τ_o = 0.5 s that is $h_{w,lb}$ = 0.9375 s, k_a^* = 0.3183 and $h^*_{w,lb}$ = 0.8727 s — the four numbers our reproduction has to hit, and does.**

WHAT THIS BUYS US

READ THE ASYMMETRY: (28a) contains no h_w at all, so no amount of headway can rescue an over-aggressive γ. That is why headway-only adaptation hits a wall at $\rho^* \approx 1.8$.

What we added — six equations, and what breaks without each

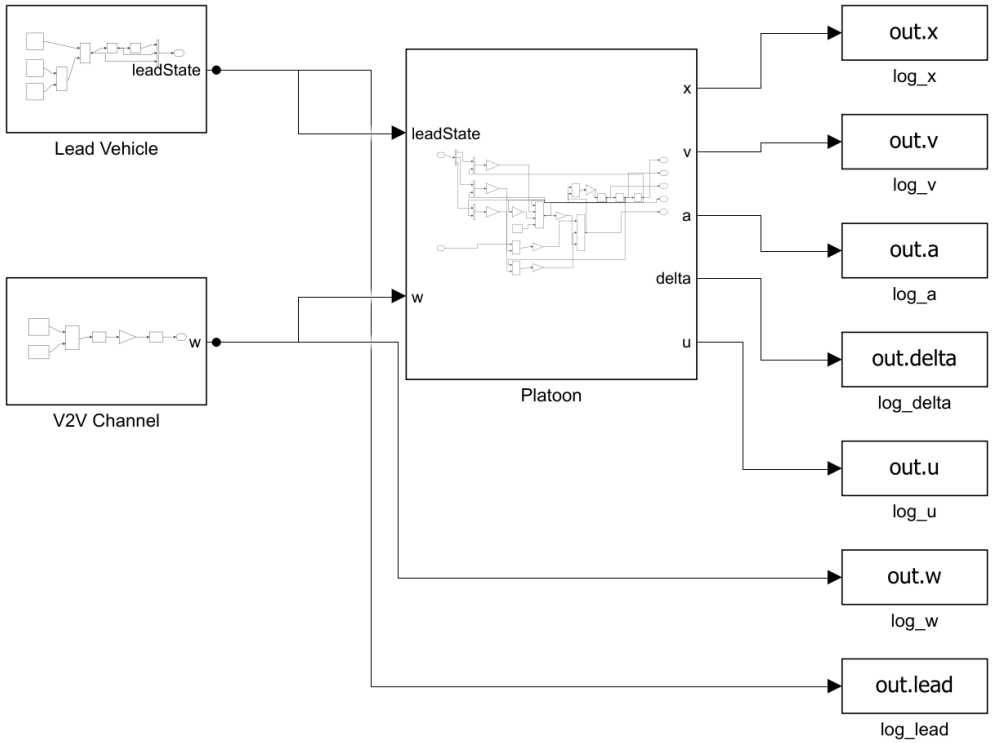
Phases 1, 3, 4 and 5 — each removes one assumption the base paper needs, and each is measured on the results pages

#	Mechanism	The equation	Take it away and…
1	**PREDICTOR** **undo the delay**	$\tilde{H}_\theta = (\tilde{k}_a\, s^2\, e^{-\theta s}P(s) + k_v\, s + k_p)/D(s)$ $P(s) = 1 + \theta\hat{\cdot}A(s)\cdot(1 - e^{-sT/2})/(T/2)$	string instability at θ ≈ 40 ms — a delay too small to think to look for. With it, the platoon survives to ≈220 ms.
2	**ESTIMATOR** **measure ρ online**	$\hat{w} = y \,/\, a_{ref}(t - \theta\!\!\!)$ $\rho\hat{} = \kappa \,/\, Q_{99}(\lvert\hat{w} - 1\rvert)$	every block below is guessing at the one parameter that matters. $\lvert w-1\rvert \le 1/\rho$ STRICTLY ⟹ any quantile errs in a known direction ⟹ $\hat{\rho}$ is conservative BY CONSTRUCTION.
3	**ADAPTER** **move the headway**	$h_i(t) = \text{table}(\rho\!\!\!)\ ,\quad \lvert\dot{h}\rvert \le 0.05\ s/s$ extra ė term $-\dot{h}\cdot v \le 1\ m/s^2$	one certified design instead of a continuum of them. The rate limit is what keeps a headway change a COMMANDED manoeuvre.
4	**SCHEDULER** **re-tune the gains**	$\gamma_{target}(\rho\!\!\!) = 0.9\cdot[1 - (1 + 1/\rho\!\!\!)^2 k_a^2]/(2\tau_0)$	no feasible design at all below ρ* ≈ 1.8, at ANY headway — because the (28a) ceiling ignores h_w.
5	**TRUST GATE** **survive a liar**	$r = \lvert y_{V2V} - a_{radar}\rvert$ $g = \text{clip}(1 - (r - r_0)/(r_1 - r_0),\ 0,\ 1)$	one spoofed link steers the car behind it into a collision. A predecessor chain has no neighbours to vote with — the radar is the only second opinion there is.
6	**CERTIFICATE** **price the risk**	$\text{exceedance}(h) = \Sigma\ P(w)\cdot 1\{\lVert\tilde{H}(\cdot;k_a\ w)\rVert_\infty > 1\}$ $h_{cc}(\varepsilon) = \min\{\ h\ :\ \text{exceedance}(h) \le \varepsilon\ \}$	you pay full worst-case price for 10^{-6} events. The law of w is known exactly (2^{16} atoms) — no need to bound, we can INTEGRATE.

Nothing here replaces the paper — it all sits on a reproduction that matches to four digits. Each row cites an ADR: D-008/016 (predictor, estimator, adapter) · D-017 (scheduler) · D-022 (trust gate) · D-023 (certificate).

Simulation — the Simulink model

matlab/models/cacc_platoon.slx — the top level, exported straight from the model



ONE model, twelve vehicles, no copy-paste

- **The whole platoon is VECTORISED to width N = 12: every signal line carries 12 lanes, so there is exactly one integrator chain and one control law in the diagram, not twelve.**

Predecessor coupling is a matrix, not wiring

- PredSel = $[I_N \mid 0]$ shifts the stacked state by one vehicle, so a_{pred} = PredSel·$[a_{lead}; a]$ — the look-ahead topology is a single gain block that can be changed without rewiring.

Three blocks, three responsibilities

Lead Vehicle — $a_0(t) = 0.5 \sin(0.1(t-10))$ over exactly one period, then constant

V2V Channel — the 16-bit multiplicative w of eq. (5)

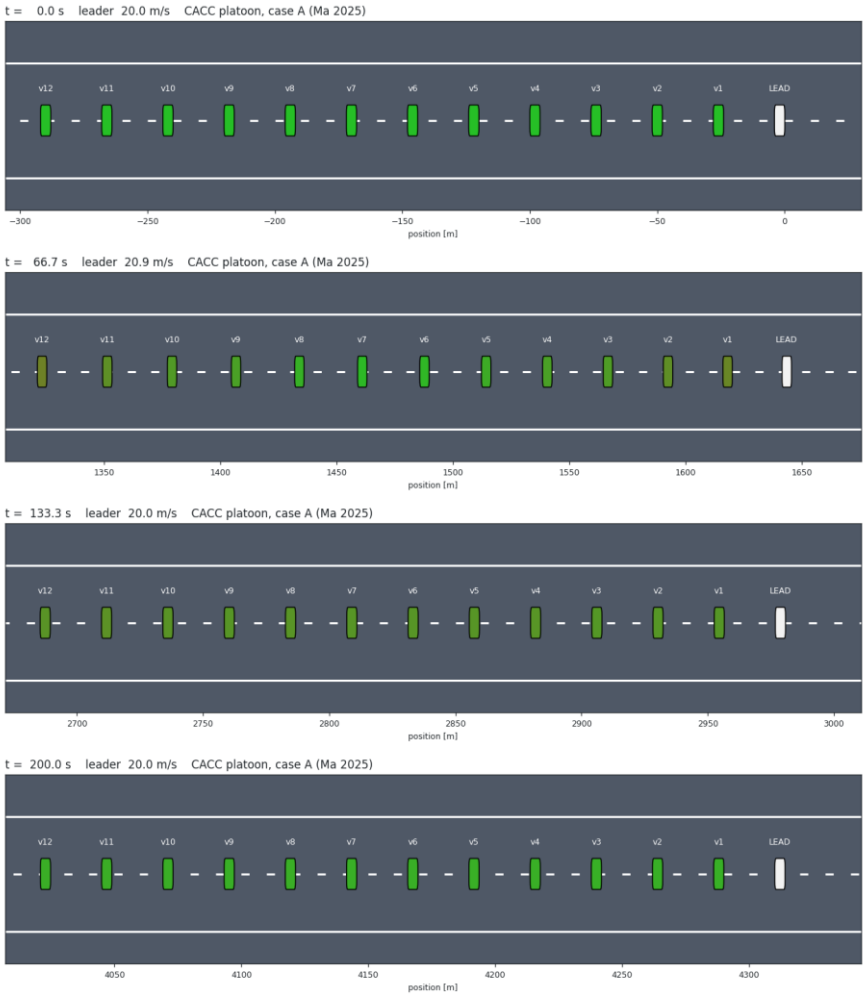
Platoon — $\delta = x - x_{pred} + h_w v + d$; $u = k_a(w \cdot a_{pred}) - k_v(v - v_{pred}) - k_p \delta$; $\dot{a} = (u-a)/\tau$

The .slx is a build artefact, not the source

- **the model is BUILT PROGRAMMATICALLY by matlab/models/build_cacc_model.m (~300 lines of add_block / add_line). The reviewable, diffable, version-controlled artefact is the .m file; the .slx is regenerated from it.**
- **That is why a third backend is worth having at all: it is an independent re-derivation of the same equations, not a re-run of the same code.**

Simulation — the platoon in motion

scenarios/ma2025_sine.yaml, case-A gains — the reproduction run, animated



Twelve followers behind one leader, top-down, to scale

- Four instants of the same 200 s run: t = 0, 67, 133, 200 s. Cars are drawn at their true 4 m length; colour encodes the spacing error (green tight, red $|e| \geq 1$ m).

What the animation is for — and what it is NOT for

- **It shows the manoeuvre is physically sensible: the wave travels backwards, gaps open and close, nothing collides.**
- **It CANNOT show string stability. A per-hop gain of 1.0035 compounds to 1.039× over 11 vehicles — invisible to the eye. That is why the verdict is always taken in the frequency domain and the animation is used only as a sanity check.**

Generated from the same core, not a separate model

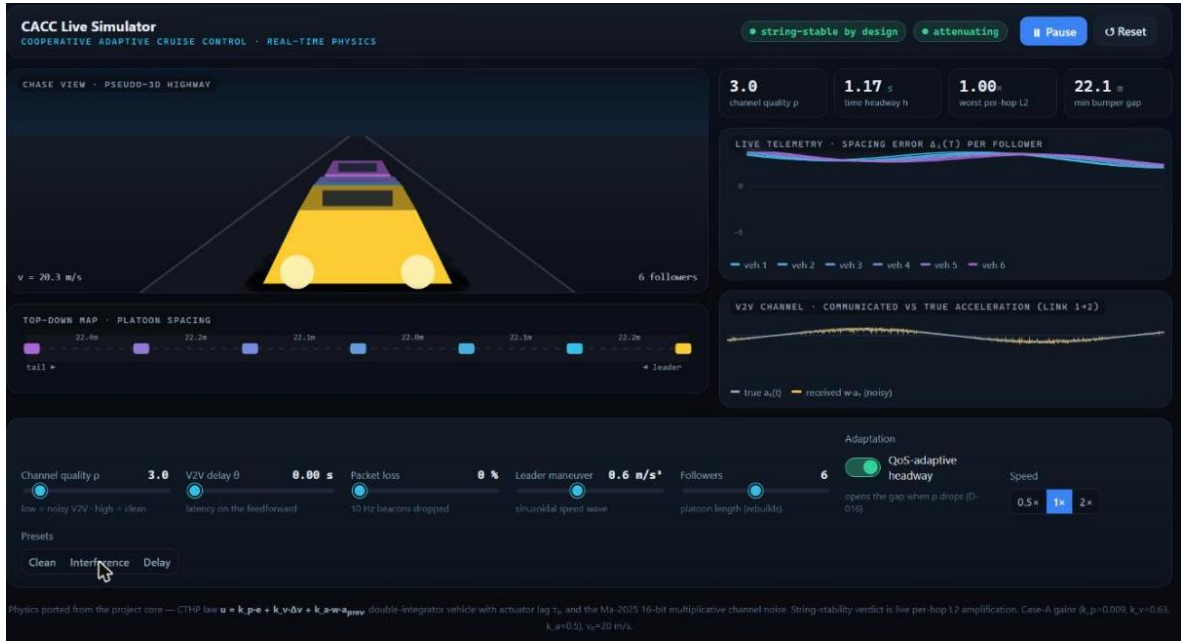
- scripts/render_platoon_video.py simulates the scenario through cacc.platoon and animates the result, so the picture and the numbers cannot disagree.

```
python scripts/make_sim_assets.py
```

- It also renders MP4/GIF at 10× time-lapse, and the ROS 2 backend drives the same scene live in rviz2 / Gazebo.

Simulation — the live, interactive demo

A live simulator, not a recording — the noise, the delay and the adaptation are all running while you watch



Real-time physics in the browser — drive it during the talk

- demo/cacc_live_sim.html — the CTHP law, the double-integrator vehicle with lag τ, and the 16-bit multiplicative channel, ported from the project core and integrated live.

What the sliders do

`channel quality ρ · V2V delay θ · packet loss · leader manoeuvre · platoon length · QoS-adaptive headway on/off`

The captured state above: the Interference preset

- **ρ dropped 5.0 → 3.0, and with QoS adaptation ENABLED the headway opened 0.95 → 1.17 s on its own; worst per-hop L_2 held at 1.00× and the minimum bumper gap stayed at 22.1 m.**
- **That is Phase 3 (D-016) happening live: the platoon detects a degraded channel it was never told about and buys back the margin in headway.**
- The V2V panel shows it directly — the received $w \cdot a_1$ trace visibly thickens into a noise band around the true $a_1(t)$.
- **Turn adaptation OFF at the same ρ and the per-hop L_2 climbs above 1 — the failure the base paper predicts, on demand.**

Results — the base paper reproduced, number by number

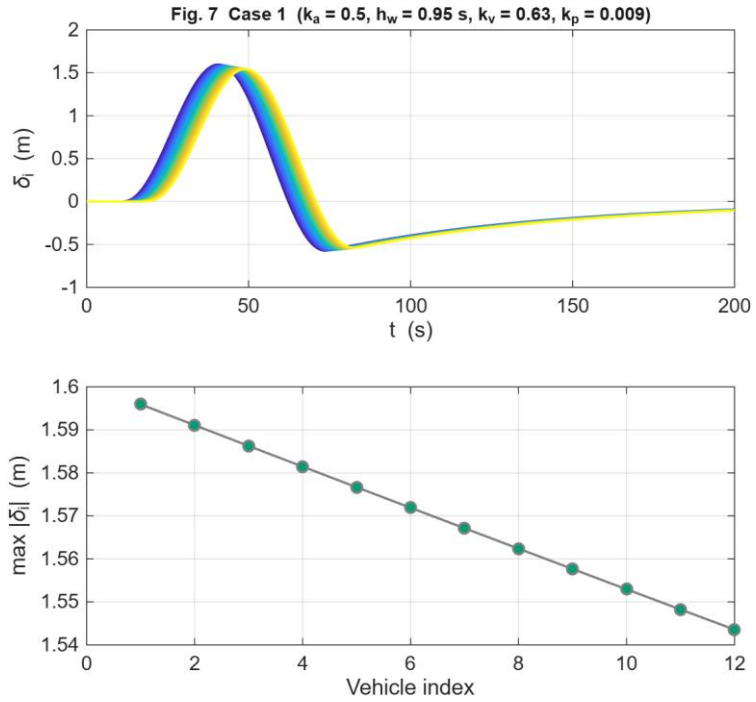
Every printed number in §IV, checked against our Simulink run (matlab/results/run_all_log.txt)

Quantity	Paper	Ours	Rel. err	Verdict
$h_{w,lb}$ ($k_a = 0.5$)	0.9375	0.937500	**0.00 %**	**EXACT**
k_a^* (eq. 18)	0.3183	0.318305	**0.00 %**	**EXACT**
$h^*_{w,lb}$ (eq. 19)	0.8727	0.872678	**0.00 %**	**EXACT**
k_a upper bound $1/(1+1/\rho)$	0.8333	0.833333	**0.00 %**	**EXACT**
Case 1 $\max\|\delta_1\| \to \max\|\delta_{12}\|$	1.593 → 1.535	1.5959 → 1.5436	**0.18 / 0.56 %**	**attenuating**
Case 2 $\max\|\delta_1\| \to \max\|\delta_{12}\|$	0.882 → 0.886	0.8821 → 0.8872	**0.02 / 0.13 %**	**AMPLIFYING**
Case 3 $\max\|\delta_1\| \to \max\|\delta_{12}\|$	0.802 → 0.798	0.9256 → 0.9208	**15.4 %**	**open item**
Case 4 platoon length, h = 0.95 / 0.88 s	345.0 / 324.0 m	345.00 / 324.00 m	**0.00 %**	**EXACT**
Case 5 $\max\|a_1\| \to \max\|a_{12}\|$, heterogeneous τ_i	≈0.49 → ≈0.455	0.4962 → 0.4523	**≈1 %**	**$\tau \in (0, \tau_o]$ holds**

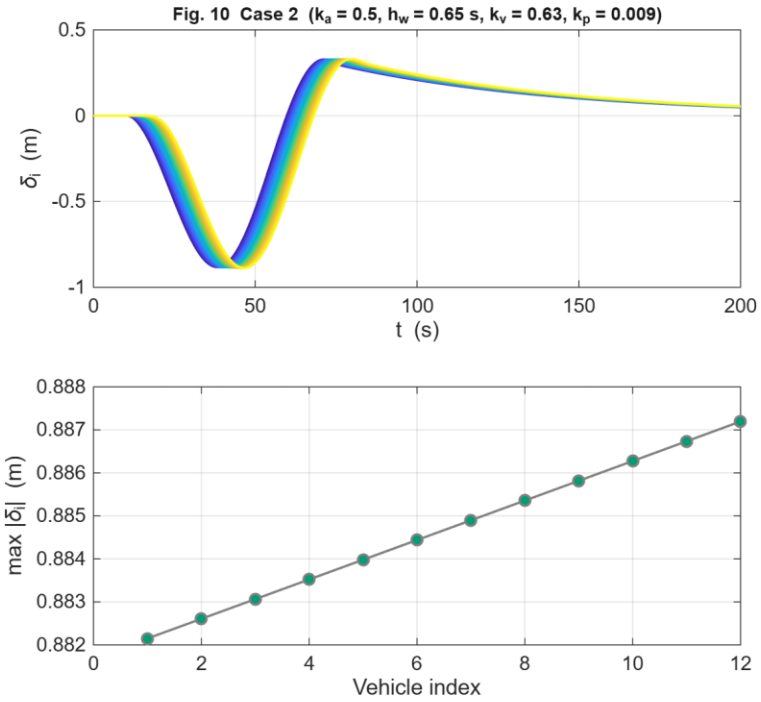
Independent cross-checks, each by a DIFFERENT route: eq. (25) residual 1.8e−12 · a 400 001-point grid search puts k_a^* at 0.318304 vs the closed form 0.318305 · Routh–Hurwitz agrees with roots(D) · Remark 4 limits recover 0.6667 / 1.0000 / 0.5000 as $\rho \to \infty$ · and a counterexample (k_a=0.5, k_v=0.70, k_p=0.009, h_w=1.50) fails (28a) yet has $\|\tilde{H}\|_\infty$ = 1.000000, confirming the conditions are SUFFICIENT, not necessary.

Results — time domain: attenuating vs amplifying

12 followers, 200 s, identical leader manoeuvre — only the headway differs



Case 1, h_w = 0.95 s > h_{lb} = 0.9375 s. Peaks SHRINK down the string: 1.5959 → 1.5436 m.

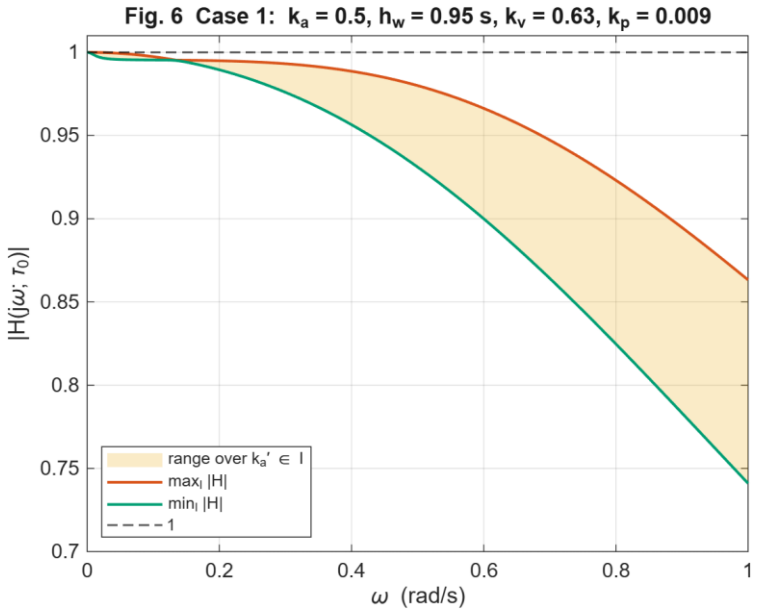


Case 2, h_w = 0.65 s < h_{lb}. Same gains, same leader. Peaks GROW: 0.8821 → 0.8872 m.

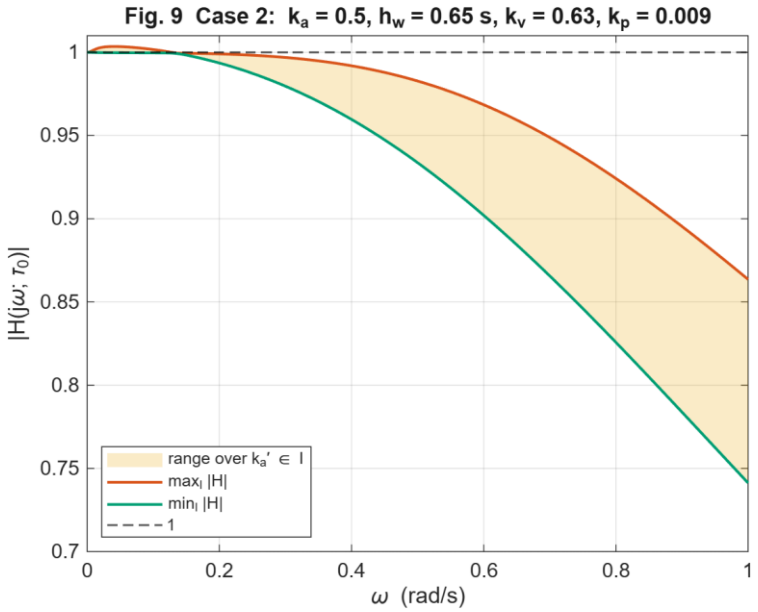
- **Case 2 is the strongest single check in the whole reproduction: the headway is violated by only 6 % and the growth is 0.5 % over 11 hops, yet we match the paper to 0.02 % INCLUDING the direction of the trend.**
- **Finding — the paper's time-domain figures come from the EQUIVALENT DETERMINISTIC system (eq. 13 / Remark 5), not from stochastic realisations. We proved this by running all three channel modes: only 'mean' mode gives the monotone max$|\delta_i|$ profiles the paper prints.**

Results — frequency domain and the design space

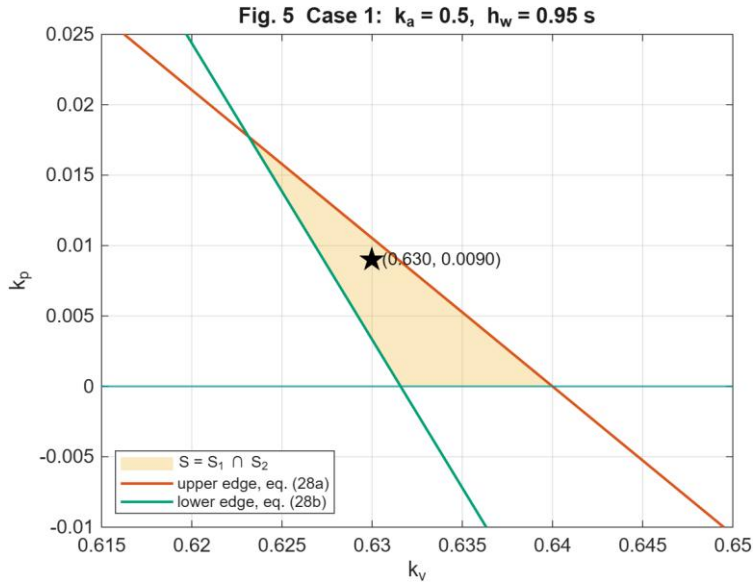
The verdict the time traces cannot give: $\|\tilde{H}\|_\infty$ over the whole uncertainty interval $I = [(1-1/\rho)k_a, (1+1/\rho)k_a]$



Case 1: sup over I and ω of $|\tilde{H}| = 1.000000$ → robustly string stable.



Case 2: 1.003500, i.e. 0.35 % above unity → string UNSTABLE.



The feasible (k_v, k_p) polygon; $a_1/a_2 = 1.0133 > 1 \Longrightarrow$ non-empty.

- **THE LOW END OF I BINDS. Case 2 gives 1.003500 at $\tilde{k}_a = 0.40$, 1.001164 at the mean, and 1.000000 at $\tilde{k}_a = 0.60$ — so evaluating only at E[w] understates the violation 3×, and evaluating at the high end declares the design SAFE.**
- **This is why hinf_worst grids $\tilde{k}_a$ across the whole interval I = [0.4, 0.6] instead of assuming which endpoint is worst.**

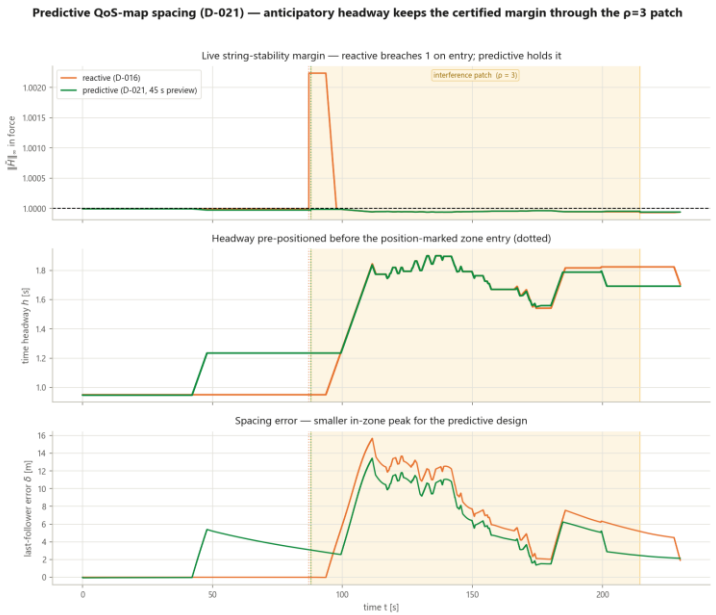
Results — the gaps we found, and what closed them

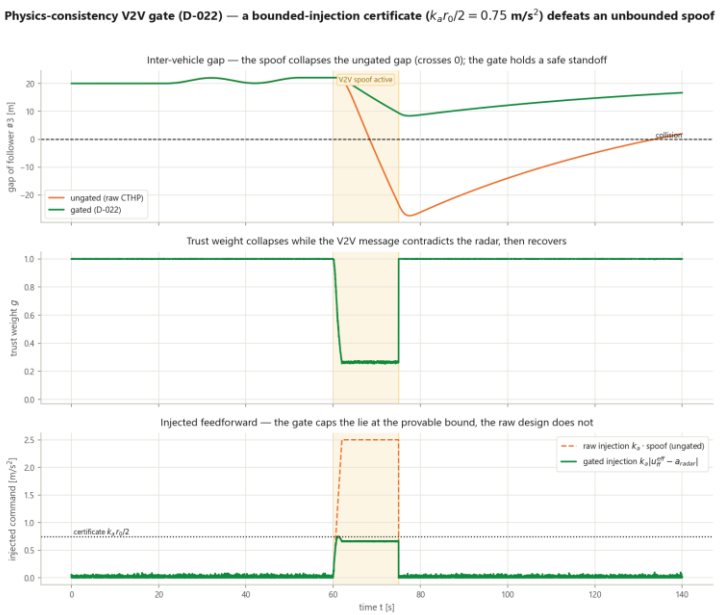
Phases 1–4 — each an assumption of the base paper, removed and measured

Phase	Gap found in the paper's design	With our extension	What it buys
1 delay & loss	at FIXED case-A gains: noiseless $h_{min}(0) = 0.789$ s, and $h_{min} \le 0.95$ s only until $\theta \approx 0.23$ s. packet loss 0 → 50 %: L_2 amplification 1.007 → 1.013	**with the timestamp predictor at $\theta = 0.15$ s: $\|\tilde{H}\|_\infty$ 1.00923 → 1.00000, and $h_{min}(\theta)$ FLATLINES**	**DELAY, not noise, is the binding constraint — the paper's own design keeps only ≈0.2 s of budget. Loss degrades gracefully.**
2 distribute	a monolithic simulator hides everything real execution adds: OS scheduling jitter, message age, per-vehicle clock skew	**one OS process per vehicle on ROS 2, wall-clock substepped RK4, radar extrapolated by message-stamp age**	**< 2 % per-hop agreement with the offline core under real jitter — the equations survive being distributed**
3 adapt	an interference zone ρ 10 → 3 → 10 with a FIXED headway leaves the platoon string-unstable for the whole time it is inside the zone	**$\hat{\rho}$ median {10.4, 3.1, 10.7} vs true {10, 3, 10}; median zone tracking error 4.75 %; headway opens and recovers**	**the channel IS estimable online from radar-vs-beacon residuals, and the estimator is conservative by construction**
4 re-tune	ρ = 2 zone, headway-only adaptation: h settles at 2.50 s with $k_v = 0.63$, in-force $\|\tilde{H}\|_\infty = 1.019$ ⟹ STILL UNSTABLE at any headway	**joint gain + headway adaptation: $h = 2.331$ s, $k_v = 0.353$, $\|\tilde{H}\|_\infty = 0.99999$**	**(28a) caps γ INDEPENDENTLY of h_w, so below $\rho^* \approx 1.8$ only re-tuning works — and it lands at a SHORTER headway**

Results — flagship A and B: prediction, and surviving a liar

Predictive QoS map (D-021) · physics-consistency V2V trust gate (D-022)





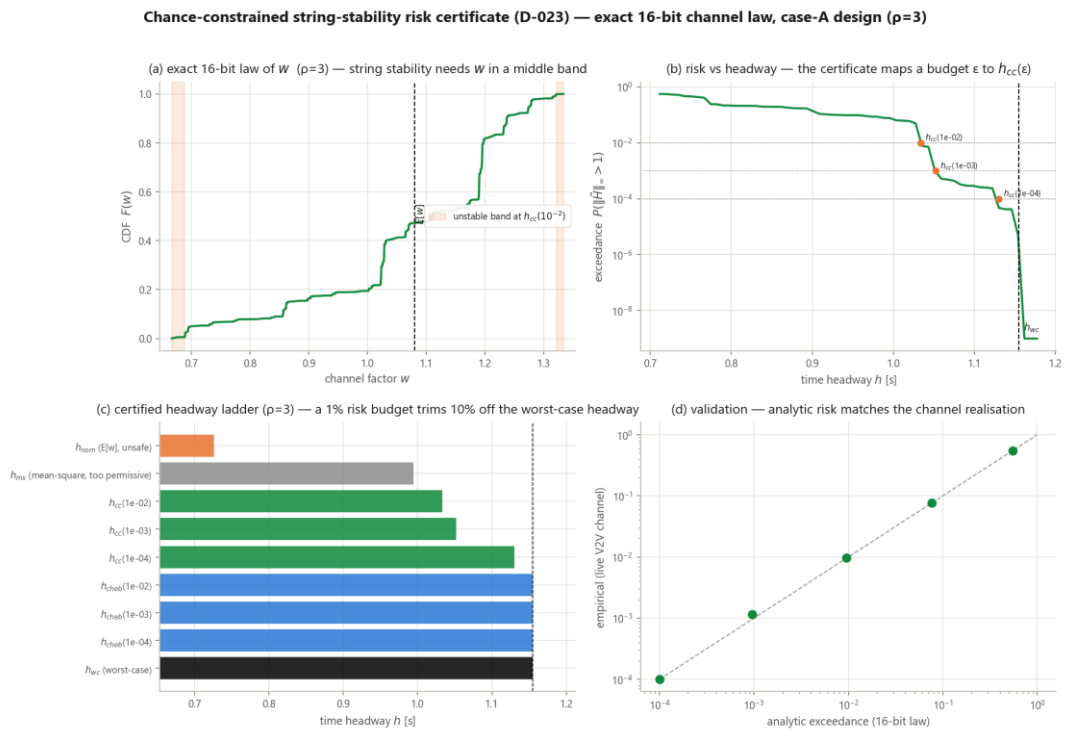
5A — REACTIVE: h = 0.95 s at zone entry, ||H̃||∞ = 1.0022, 6.9 % of in-zone time string unstable, peak error 17.23 m. PREDICTIVE (45 s look-ahead on the ρ(x) map): h = 1.234 s already at entry, ||H̃||∞ = 1.0000, 0.0 % unstable, peak error 14.23 m.

5B — UNGATED under a +5 m/s² bias injected for 15 s: minimum gap −27.4 m, i.e. a COLLISION, peak error 50.5 m. GATED: minimum gap +8.37 m, no collision, peak error 14.24 m, injection capped, minimum trust weight 0.254.

- **5A — the platoon is already at the right headway BEFORE it enters the degraded zone, so the instability is ELIMINATED rather than merely shortened. Reacting to a channel you have already entered is late by one adaptation time constant.**
- **5B — a predecessor-following chain has no neighbours to vote with, so consensus filtering is unavailable. The gate uses the fact that the same quantity arrives TWICE — over the radio and off the radar — and trusts the radio only as far as physics allows.**

Results — flagship C: a binary verdict replaced by a priced risk

Chance-constrained string-stability certificate (D-023) — exact 16-bit channel law, case-A design



The law of w is known EXACTLY

- **2^{16} = 65 536 atoms, obtained by convolving the 16 Bernoulli bit laws. So there is no need to BOUND — we can INTEGRATE.**

At ρ = 3, case-A gains

```
worst case          h = 1.1544 s   (exceedance 0)
chance-constrained  h = 1.0334 s   (exceedance 0.0095)
chance-constrained  h = 1.0522 s   (exceedance 0.0010)
mean-square         h = 0.9944 s   (exceedance 0.076)
```

- **–10.5 % headway for a calibrated 1-in-100 risk budget; the analytic and Monte-Carlo exceedances agree to 3–4 decimals.**
- **The mean-square criterion is reported but REJECTED: 7.6 % exceedance at ρ = 3 is far too permissive to certify with.**

Worst-case design pays full price for events of probability 10^{-6}. This states the price instead — a regulator sets ε, and the method returns the headway.

The verdict, and what this project concludes

Monte-Carlo certification harness (D-018) — and the six things this project claims

CERTIFIED

the harness verdict

4 scenario families × seeds;
11 numeric checks, ALL PASS

0.00 %

reproduction error

on all four of the paper's closed forms

87

unit tests green

the paper's printed numbers pinned as assertions

1e–4

Python ↔ Simulink

an independent re-derivation, not a re-run of the code

- **Three criteria per certification run — theory (in-force $\|\tilde{H}\|_\infty \le 1$), time domain (per-hop L_2), and SAFETY: minimum gap on the NONLINEAR simulation, because actuator saturation sits outside the linear envelope $\|\tilde{H}\|_\infty$ describes.**
- **1. The base paper reproduces EXACTLY — the credibility anchor for everything else.**
- **2. Its guarantee is real but THIN, and it is DELAY, not noise, that consumes the margin.**
- **3. A fixed design cannot survive a changing channel — and headway alone cannot save it below $\rho^* \approx 1.8$.**
- **4. The channel IS estimable online, and the estimate is conservative by construction.**
- **5. A calibrated risk budget is cheaper AND more honest than a worst case.**
- **6. Reproducibility is the deliverable — 87 pinned tests, seeded noise, three independent backends.**